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Recommended reading and resources

T.V.N. (Vid) Persaud, MD, PhD, DSc, FRCPath (Lond.), FAAA; Mark G. Torchia, MSc, PhD

[The Developing Human](#), 12th ed. Chapter 12, pg. 225-264

Images:

Gary C. Schoenwolf, PhD; Steven B. Bleyl, MD, PhD; Philip R. Brauer, PhD; Philippa H. Francis-West, PhD

[Larsen's Human Embryology](#), 6th ed. Chapter 16, pg. 386-418

Other resources:

Urogenital System. In: Sadler TW. eds. *Sadler*15. Lippincott Williams & Wilkins, a Wolters Kluwer business; 2024. Accessed July 18, 2025.

<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?bookid=3221§ionid=253323054>

Objectives

SOM.MKII.BPM2.1.ER.1.ANAT.1081	Describe the anatomical location, mechanism and clinical consequences in the development of hydrocele testis.
SOM.MKII.BPM2.1.ER.1.ANAT.1082	Describe defects in testicular descent including cryptorchidism and ectopic testes.
SOM.MKII.BPM2.1.ER.1.ANAT.1088	Explain the developmental mechanism responsible for the formation of hypospadias and epispadias.
SOM.MKII.BPM2.1.ER.1.ANAT.1089	Describe the identification and clinical significance of disorders of sex development (DSD's)
SOM.MKII.BPM2.1.ER.1.ANAT.1090	Describe the consequence of decreased levels of dihydrotestosterone on development of male characteristics.
SOM.MKII.BPM2.1.ER.1.ANAT.1091	Describe the cause and clinical picture of androgen insensitivity syndrome.
SOM.MKII.BPM2.1.ER.1.ANAT.1085	Describe the development of birth defects of the female genital tract including a double uterus, a unicornuate uterus, a bicornuate uterus vaginal agenesis and imperforate hymen.
SOM.MKII.BPM2.1.ER.1.ANAT.1086	Describe the anatomical location, mechanism and clinical sequences of canal of Nuck hernias.
SOM.MKII.BPM2.1.ER.1.ANAT.1092	Describe the formation and clinical significance of Gartner duct cysts.

Lecture outline

Disorders of sex development (DSD) – general concepts

- Ovotesticular DSD

Disorders of sex development when Y-Chromosome is present (46, XY DSD)

- General concepts
- Hypospadias
- Specific examples
 - 5 α -reductase deficiency
 - Androgen Insensitivity syndrome

Other developmental disruptions in the presence of Y-Chromosome

- External genital organs
 - Epispadias
- Developmental issues concerning the gonads
 - Failure of the testes to descend into the scrotum and its consequences
 - Ectopic locations of testes
 - Failure or incomplete obliteration of the processes vaginalis

Disorders of sex development when Y-Chromosome is absent

Gartner cysts

- Canal of nuck

Disorders of sex development - DSD

Disorders of sex development is an inclusive term that refers to any problem where the genital organ is atypical in relation to chromosomes or gonads

Effect of chromosome configuration (karyotype) on development of the reproductive system

- For development of male (or masculine) phenotype, the Y chromosome is required
- For development of female (or feminine) phenotype, the Y chromosome must be ABSENT, and two X chromosomes are required to ensure the ovaries develop normally
- Additional X chromosomes, as seen in some syndromes, do not affect the development of the gonads or external genital organs in either sex

While many DSD's are caused by alterations in the sex chromosomes most genital system malformations have been shown to result from autosomal gene alterations

Disorders of sex development – Ovotesticular DSD

- Presence of both ovarian and testicular tissue, often in a single organ (ovotesticular organs)
 - May be an ovary or ovotesticular organ on one side and a testes on the other
- Genetically may be chromosomal males (46, XY) or females (46, XX) or they may have 3 sex chromosomes (47, XXY) or be mosaic where they have (46, XY) in some cells and (46, XX) in others
 - Most frequently (70%) occurs in 46, XX individuals and may be explained by a segment of the Y chromosome translocated onto one of the X. Inactivation of the X in some cells may then cause the mosaicism
 - 46, XY individuals are not easily explained, only 15% of these individuals have SRY gene mutations
- External genitalia can be ambiguous but are most frequently masculinized
- Ovulation may occur and sperm production has been observed but successful reproduction has not been documented

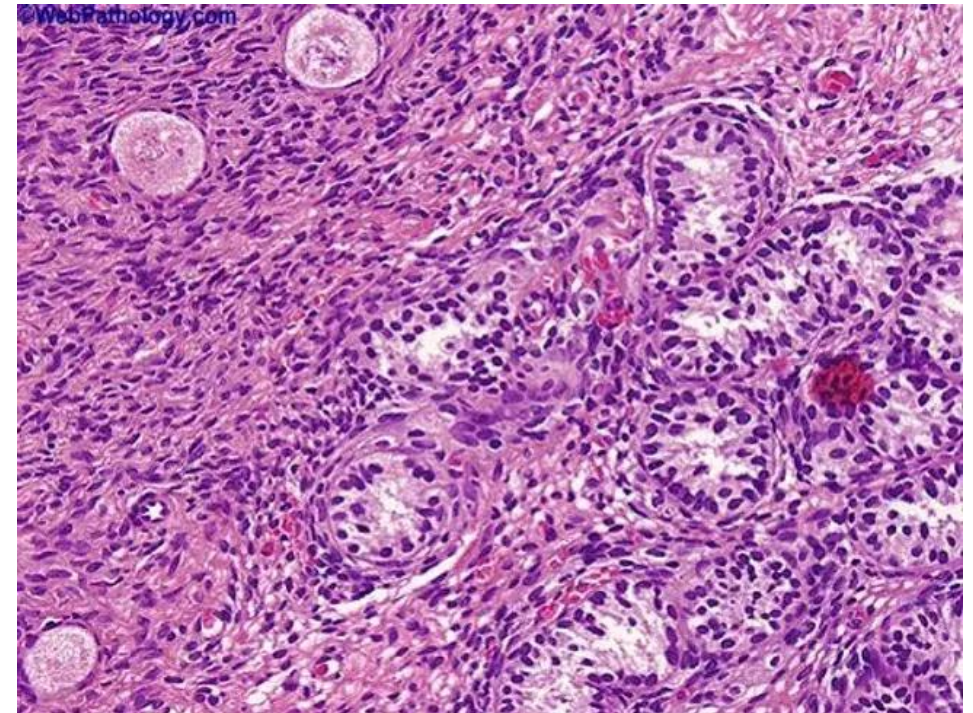


Image taken from: <https://www.webpathology.com/images/genitourinary/testis/congenital-anomalies-and-infertility/36394>

Disorders of sex development when **Y-Chromosome** is present (46, XY DSD)

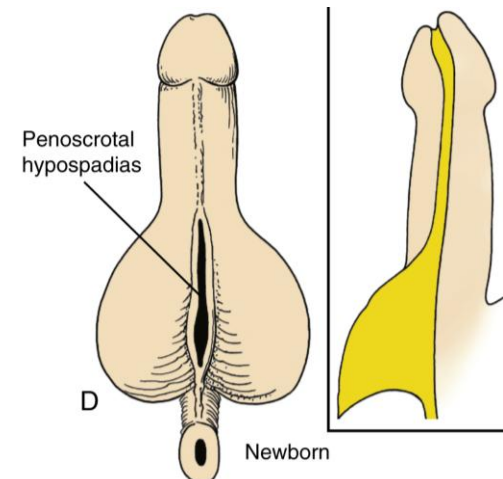
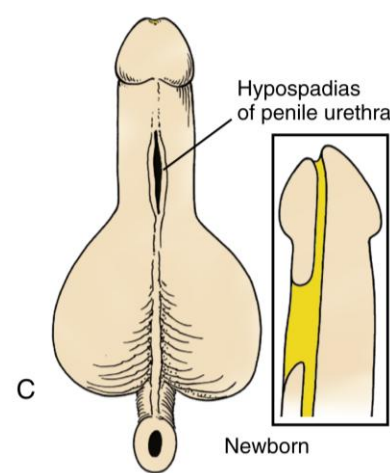
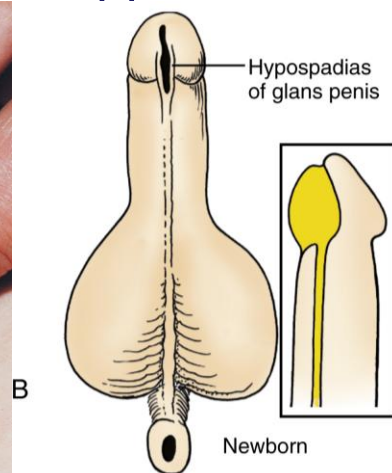
- General concepts
- Hypospadias
- Specific examples
 - 5 α -reductase deficiency
 - Androgen Insensitivity syndrome

Disorders of sex development - 46,**XY** DSD

- Due to a deficiency in androgen action
- Causes feminization of the external genital organs
- What is affected is determined by the hormone that is deficient
 - May be the result of **testosterone absence/deficiency**
 - Mutations in enzymes required for testosterone synthesis
 - Mesonephric ducts don't differentiate
 - Testes don't descend
 - External genital organs are feminized
 - Paramesonephric ducts degenerate because of AMH
- Most common feature is hypospadias (not restricted to 46, XY DSD)
- May be the result of 5 α -reductase deficiency or inadequacy
- May result from the absent or disabled androgen receptors – Androgen insensitivity Syndrome

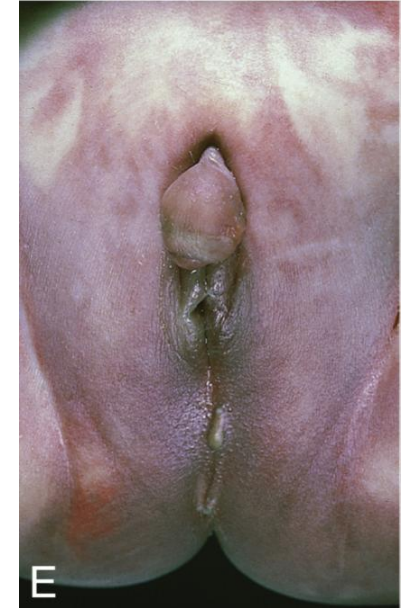
In the presence of **Y-Chromosome** Urethral abnormalities - Hypospadias

- External urethral orifice located along the ventral surface of the penis rather than the tip of the glans
- Cause is dependent largely on location:
 - Glans (glanular) – defective cannulization of the distal glans plate
 - Body of the penis (penile) – failure of the urethral folds to fuse completely
 - Penoscrotal (perineal) – failure of labioscrotal folds and urethral folds to fuse
- The most severe type of hypospadias causes complete non-fusion of the urethral folds and the labioscrotal folds and restricted growth of the penis
 - Genitals appear female at birth



In the presence of **Y-Chromosome** **46,XY DSD, 5 α -reductase deficiency**

- Causes severe penoscrotal hypospadias
- Feminized genital organs at birth
- Testes are normal but remain undescended (in the inguinal canal)
- Ductal system is normal as the testosterone levels and AMH remain unaffected
- The increase in testosterone production at puberty may cause external genital differentiation and fusion of the labioscrotal folds
 - Causes differentiation and maturation of accessory genital glands
 - These individuals may be fertile and produce offspring

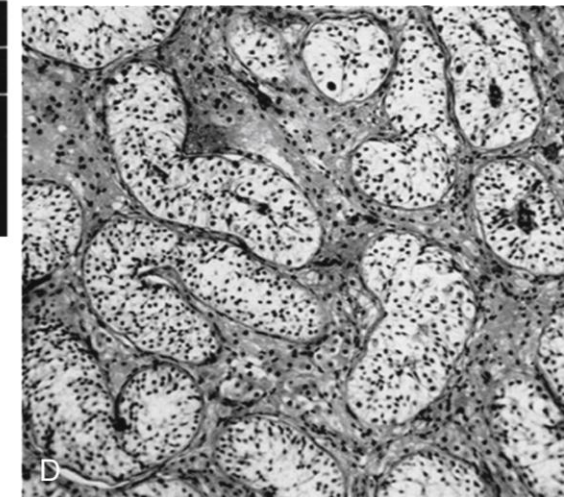
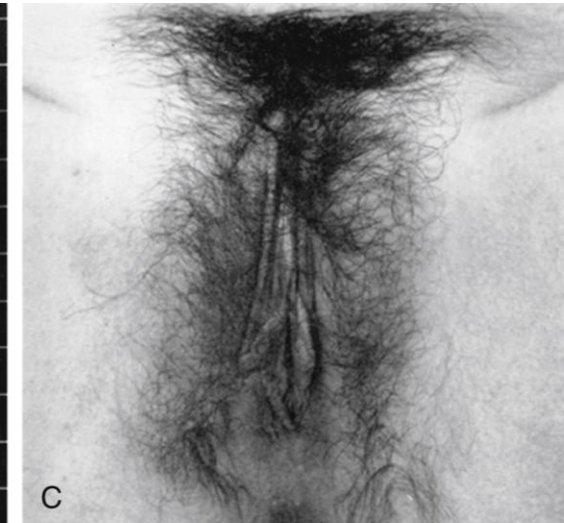


In the presence of **Y-Chromosome** 46,XY DSD – Androgen Insensitivity Syndrome (AIS)

- Androgen receptors are absent or disabled
- Normal or high levels of testosterone produced by the fetus
- Two types which determine the presentation
- Complete (CAIS)
 - Testes are present but remain in the inguinal canal in most cases
 - Spermatogenesis does not occur
 - High risk of malignancy
 - Paramesonephric ducts regress
 - Blind ending vagina may form
 - Phenotype at birth is feminized and development of female secondary characteristics occur at puberty
- Mild or partial (MAIS/PAIS)
 - Phenotype can be a range from completely feminized, ambiguous or even male (presenting with infertility)
 - Testes are located in the inguinal region



17-year-old

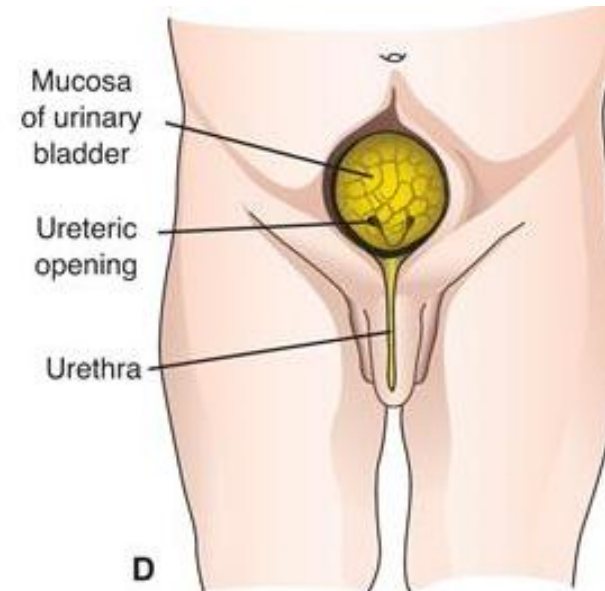
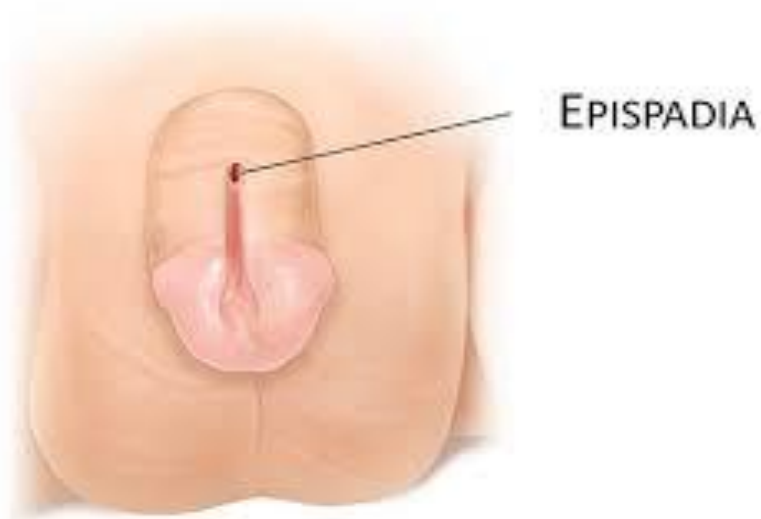


Other developmental disruptions in the presence of **Y-Chromosome**

- External genital organs
 - Epispadias
- Developmental issues concerning the gonads
 - Failure of the testes to descend into the scrotum and its consequences
 - Ectopic locations of testes
 - Failure or incomplete obliteration of the processes vaginalis

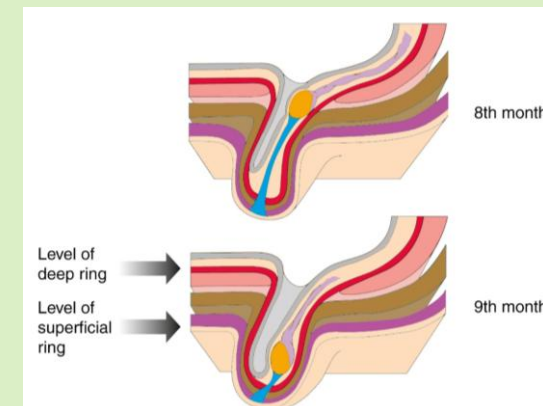
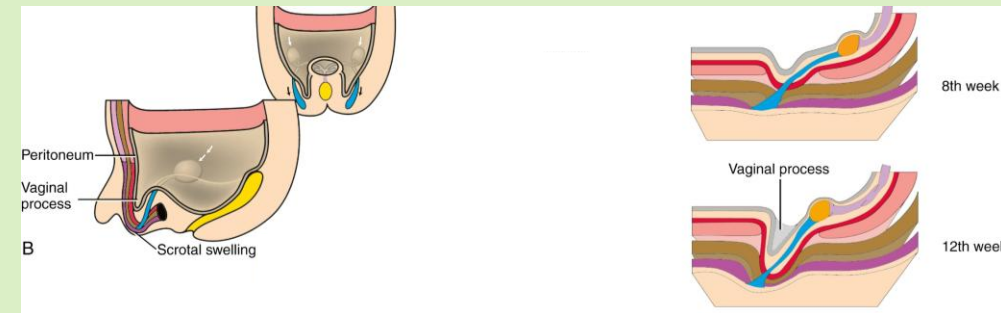
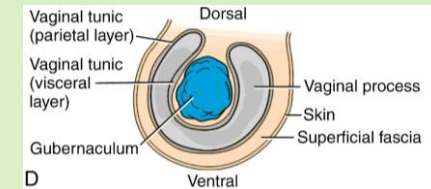
In the presence of **Y-Chromosome** Urethral abnormalities - Epispadias

- Urethra opens on the dorsal surface of penis
- Results from inadequate ectodermal-mesenchyme interactions during development of the genital tubercle
- Genital tubercle develops more dorsally and when the urogenital sinus ruptures to form the urethral groove it does so dorsally
- Often associated with exstrophy of the bladder
- Affects 1:30000



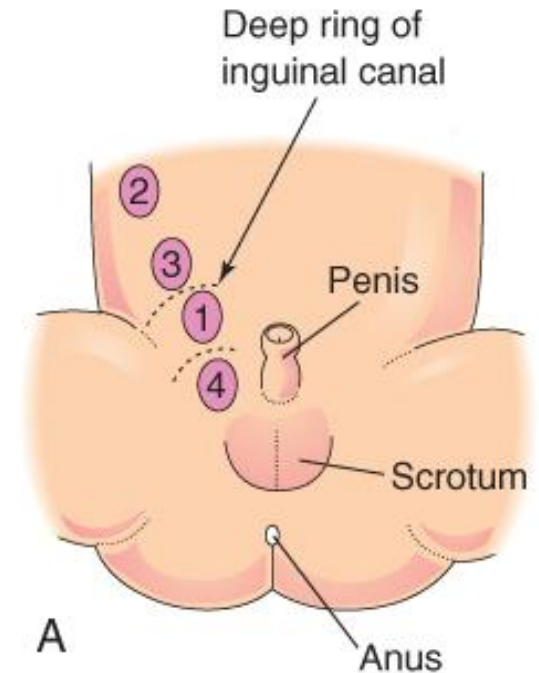
Development of the inguinal canal and testicular descent

- An evagination of the peritoneum, the **processus vaginalis**, develops around the ventral aspect of the gubernaculum
 - It hollows out the inguinal canals and the labioscrotal swellings
 - It grows inferiorly pushing out a sock-like evagination that consist of several layers of the abdominal wall
 - Transversalis fascia
 - Internal oblique muscle and fascia
 - External oblique muscle
- The **gubernaculum** shortens and pulls the testes inferiorly along the abdominal wall, from 3rd to 7th month the testes are situated just cranial to the deep ring
- From the 7th month to the 9th month, the gubernaculum swells (**causing a widening of the inguinal canal**) and shortens more, guiding the testes through the canal (androgen dependent) and into the scrotum
- The cranial suspensory ligament regresses



In the presence of **Y-Chromosome** – Failure of the testes to descend - **Cryptorchidism**

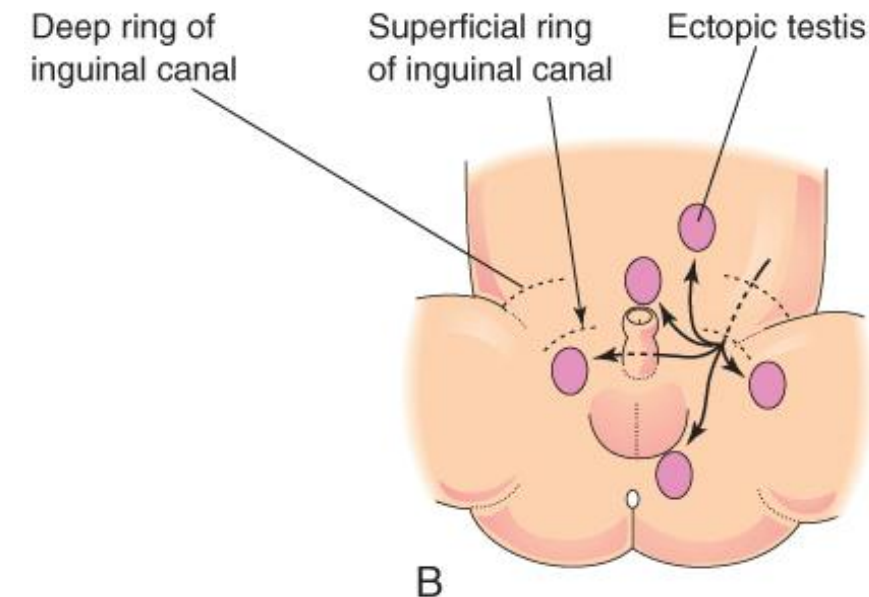
- **Cryptorchidism** means hidden testes
- May be located anywhere along the descent pathway but most frequently in the **inguinal canal**
- Many infants are born with undescended testes but usually by 3 months it self-corrects
- Testes may remain undescended for many reasons and are associated with a variety of DSD
 - The causes may be due to androgen insufficiency or other molecular factors but are largely unknown
- It can be surgically relocated by performing an **orchiopexy** and should be done if undescended for more than 6 months
- Failure to mature and subsequent sterility is common
- Have an increased risk of testicular cancer because the higher (by 3 to 5°C) abdominal temperature leads to less availability of spermatogonia type A and higher malignant transformation of the germ cells



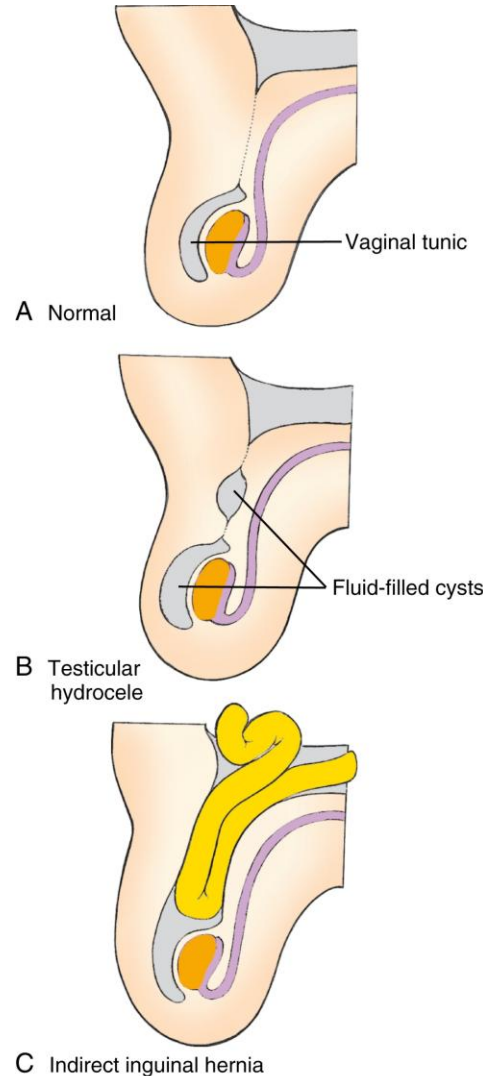
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In the presence of **Y-Chromosome** – ectopic testes

- **Very rare**
- Testes may deviate from their path after they descend **through the inguinal canal**
- They may settle in various locations
 - **Interstitial (external to the aponeurosis of the external oblique muscle) – most frequent**
 - In the proximal part of the medial thigh
 - Dorsal to the penis
 - On the opposite side (crossed ectopia)
- Caused by an abnormal placement of the gubernaculum



In the presence of **Y-Chromosome** - Failure or inadequate obliteration of the processus vaginalis



- If the obliteration does not occur it leaves a direct communication between the peritoneal cavity and the scrotum
- This may lead to hydrocele (accumulation of fluid in the scrotum) or to herniation of intestines through the internal inguinal ring = **congenital inguinal hernia**
- Once the processus vaginalis have obliterated it forms a saclike structure ventral to the testes which can be substantial in size during infancy = **processus vaginalis**
- Normally it is empty and flattened but serous fluid may accumulate forming a cyst = **hydrocele**
- The signs of the two conditions may appear the same
 - Transillumination test helps to differentiate

Disorders of sex development - DSD

Disorder of sex development is an inclusive term that refers to any problem where the genital organ is atypical in relation to chromosomes or gonads

Effect of chromosome configuration (karyotype) on development of the reproductive system

- For development of male (or masculine) phenotype, the Y chromosome is required
- For development of female (or feminine) phenotype, the Y chromosome must be ABSENT, and two X chromosomes are required to ensure the ovaries develop normally
- Additional X chromosomes, as seen in some syndromes, do not affect the development of the gonads or external genital organs in either sex
- There are clinical syndromes associated with multiple X chromosomes however their effect is more related to other systems in the body

While many DSD's are caused by alterations in the sex chromosomes most genital system malformations have been shown to result from autosomal gene alterations

Developmental issues when Y-Chromosome is absent

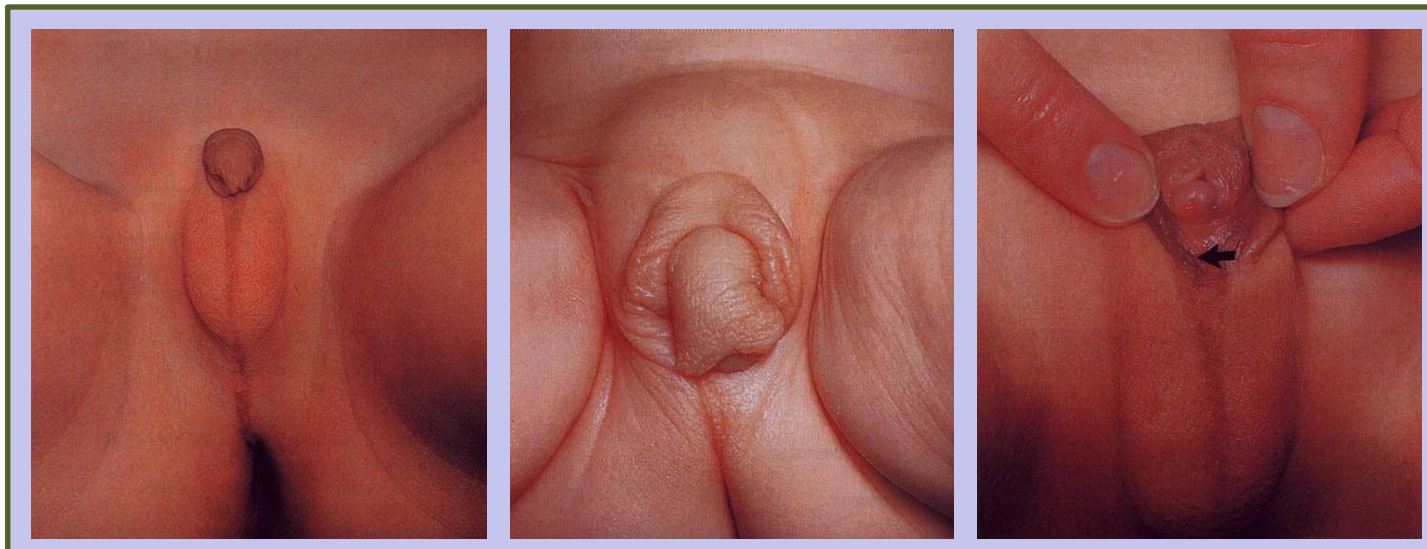
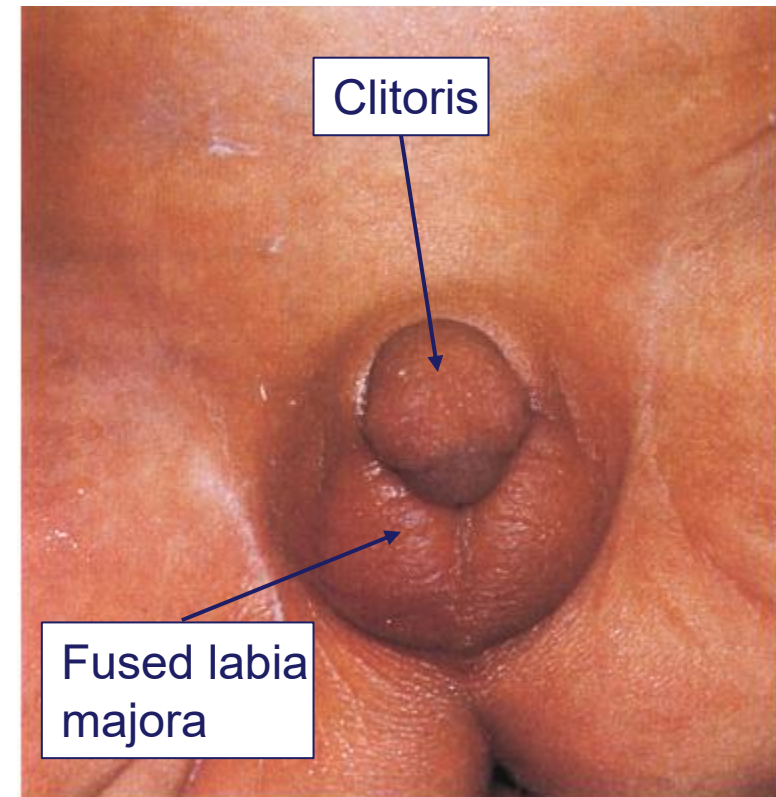
Disorders of sex development when Y-Chromosome is absent – DSD 46, XX

Developmental issues when Y-Chromosome is absent

- Developmental issues related to female ducts
- Gartner cysts
- Canal of Nuck

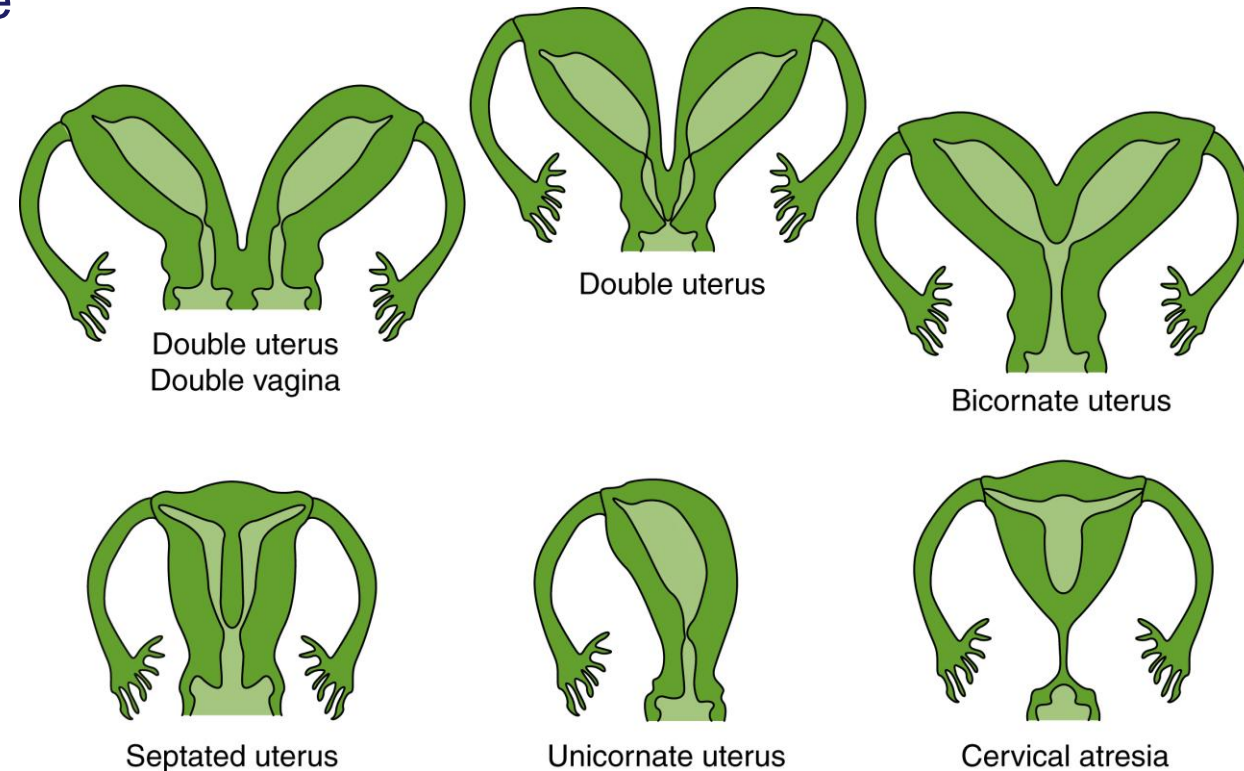
Disorders of sex development - 46,**XX** DSD

- An excess in virilizing androgens due to suprarenal gland hyperplasia
- Causes ambiguity and masculinization of the external genital organs
- Enlarges clitoris
- Fusion of labia
- But normal development of vagina and uterus
- Ovaries are present and normal



In the **absence** of Y-Chromosome Developmental issues related to female ducts

- Rare only 1% of fertile women and 3% of women who experience repeated abortions
- Most women can conceive but they have higher rates of miscarriages, premature births and difficult births
- Related to abnormal fusion or regression of the caudal portion of the paramesonephric ducts
- Incomplete fusion = Double uterus, bicornate
- Failure of uterine septum to regress = septated uterus or cervical atresia may also cause bicornate
- Complete regression of one paramesonephric duct = Unicornate uterus



In the **absence** of Y-Chromosome Female genital tract defects

Vaginal agenesis

- Failure of the sinovaginal bulbs to develop from the vaginal plate
- Normally associated with the absence of the uterus (The developing uterus induces the formation of the sinovaginal bulbs)

Vaginal atresia

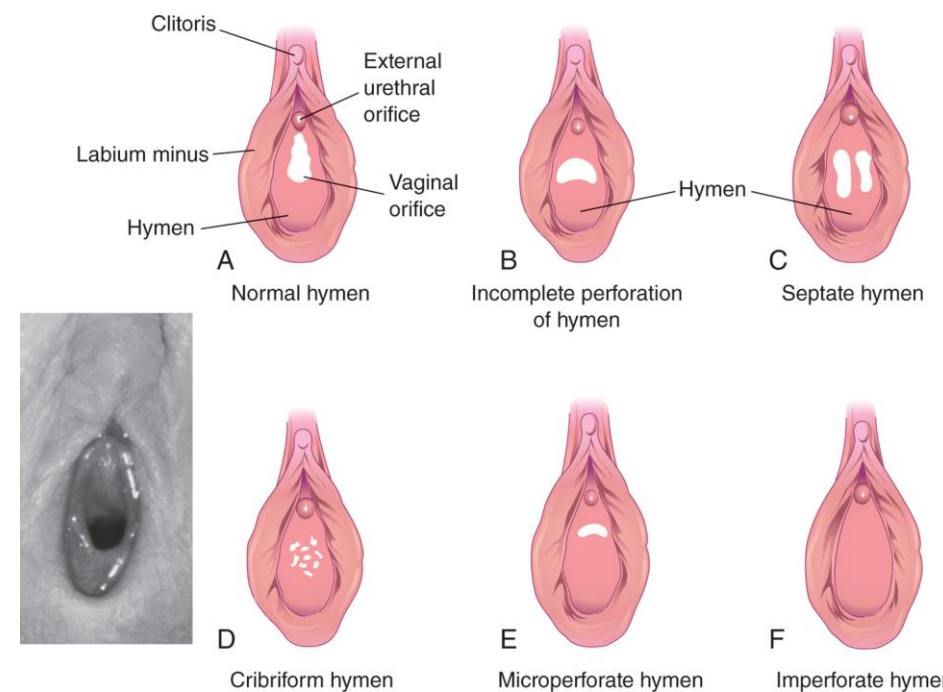
- Failure of canalization of the vaginal plate

Imperforate hymen

- Failure of the inferior end of vaginal plate to perforate

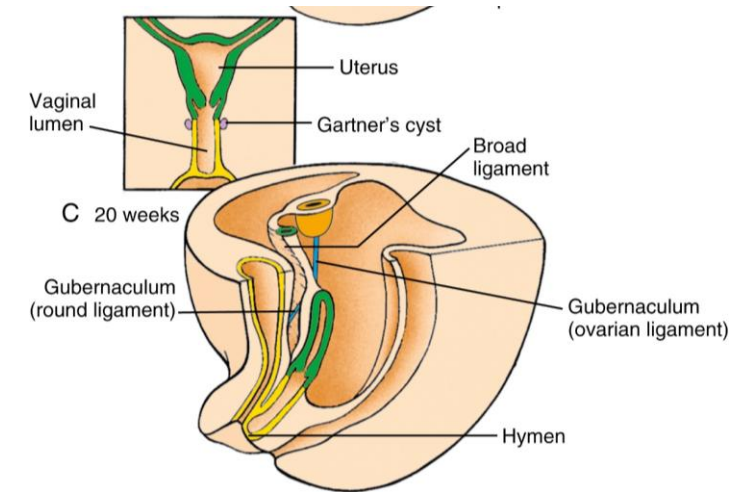
Main presentation is a pubertal girl with
amenorrhea

If the uterus is present (Vaginal atresia or
imperforate hymen) then 2+ months of lower
abdominal cramping and pelvic pain are typical
symptoms



In the **absence** of Y-Chromosome Gartner's (Duct) Cyst

- **Gartner's cysts** develop from a portion of the mesonephric duct that remains along the uterine and vaginal walls
- Benign cyst that originates from the Gartner's duct, which is a vestigial remnant of the mesonephric duct (wolffian duct)
- They are typically small and asymptomatic but may become enlarged affecting adjacent structures
- Generally located above the lower limit of the pubic symphysis



In the **absence** of Y-Chromosome Canal of Nuck hernias

- Canal of Nuck is the name given when the processus vaginalis is not obliterated in females
- Round ligament (remnant of gubernaculum) passes through the inguinal canal to insert into the labia majora
- In its path it also carries with it a portion of the peritoneum **Processes vaginalis**
- If patent can allow for development of inguinal hernias or hydroceles



Chan D, Kwon JK, Lagomarsino EM, Veltkamp JG, Yang MS, Pfeifer CM. Canal of Nuck hernias. *Acta Radiologica Open*. 2019;8(12). doi:[10.1177/2058460119889867](https://doi.org/10.1177/2058460119889867)

